Section – A Concept Application

S1 . Building a new office workstation

- **Choose the CPU first** – Check the CPU socket type, such as AM5 or LGA1700.
- **Select a compatible motherboard** – The motherboard must have the same CPU socket and support the CPU generation.
- **Choose RAM** – Check the motherboard's supported RAM type, such as DDR4 or DDR5, maximum capacity, and supported speed.
- **Choose storage** – Check whether the motherboard supports the required storage type, such as SATA SSD/HDD or M.2 NVMe.
- **Check the motherboard specifications** – Confirm that there are enough RAM slots, M.2 slots, SATA ports, and other required connections.

S2. HDD vs SSD vs NVMe

HDD

- **Uses spinning magnetic disks.**
- **Has moving parts.**
- **Slower than SSDs.**
- **Usually cheaper for large storage capacities.**
- **Can make noise and use more power.**

- **SSD**
  - **Uses flash memory and has no moving parts.**
  - **Much faster and quieter than an HDD.**
  - **More resistant to physical shock.**
  - **Usually uses a SATA connection.**
- **NVMe SSD**
  - **Also uses flash memory.**
  - **Usually connects through an M.2 slot using PCIe.**
  - **Much faster than a SATA SSD.**
  - **Has very low access times and high data-transfer speeds.**

S3. GPU and cooling system

- **The GPU (Graphics Processing Unit) processes images, videos, and graphics. For mildly graphics-intensive design software, a suitable GPU can make the software run more smoothly and improve rendering and display performance.**
- **When the GPU is working for several hours, it produces heat. If the temperature becomes too high, the GPU may reduce its speed automatically. This is called thermal throttling, and it can reduce performance.**

The cooling system removes heat from the GPU, CPU, and other components.

A good cooling system may include:

- **CPU heatsink and fan**
- **GPU heatsink and fans**
- **Case fans**
- **Good airflow through the case**
- **Thermal paste between the CPU and heatsink**

Good cooling keeps temperatures under control, prevents overheating, and helps the workstation remain stable and reliable during long periods of use.

S4. Why hardware compatibility matters

Hardware compatibility is important because computer components must be able to communicate and physically connect with each other.

For example:

- **The CPU must fit the motherboard's socket.**
- **RAM must be the correct type for the motherboard.**
- **The motherboard must support the CPU.**
- **Storage must use a supported connection such as SATA**
- **The PSU must provide the correct power connectors and enough power.**
- **The case must have enough space for the motherboard and other components.**

Problems caused by incompatible components can include:

- **The computer may not boot.**
- **The CPU may not work with the motherboard.**
- **RAM may not be recognised.**
- **Storage may not be detected.**
- **Components may not physically fit.**
- **The system may crash or become unstable.**
- **The PSU may not provide enough power.**
- **In the worst case, incorrect power connections can damage hardware.**

Therefore, I would check the specifications of all components before assembling the PC.

S5. Choosing and connecting the PSU

First, I would calculate the approximate power requirement of the complete system, including:

- **CPU**
- **GPU**
- **Motherboard**
- **RAM**
- **Storage drives**
- **Cooling fans**
- **Other connected devices**

I would then choose a PSU with enough wattage and some extra capacity. For example, if the system is expected to use around 400 W, I would not choose a 400 W PSU; I would choose a higher-rated unit with suitable headroom.

I would also check that the PSU has the correct:

- **24-pin motherboard connector**
- **CPU/EPS power connector**
- **PCIe power connectors for the GPU, if required**
- **SATA power connectors for storage**

Safety checks before powering on:

1. **Turn off the PSU and unplug it from the wall.**
2. **Make sure the PSU is suitable for the system.**
3. **Check that all connectors are fully inserted.**
4. **Do not force connectors into the wrong sockets.**
5. **Check that motherboard screws and other components are installed correctly.**
6. **Make sure there are no loose screws or metal objects inside the case.**
7. **Check that CPU and GPU cooling fans are connected.**
8. **Make sure the CPU cooler is properly installed.**
9. **Check that cables are clear of fans.**
10. **Do a final visual inspection before switching the PSU on.**

Only after these checks would I connect the power cable and perform the first boot.

S6. Checking the workstation before handing it over

- **Before giving the workstation to the client, I would make sure everything is working properly.**
- **First, I would turn the computer on and check that it boots normally. I would check that the RAM, CPU and storage are detected.**
- **I would install the required drivers, operating system updates and software.**
- **I would then test the keyboard, mouse, monitor, USB ports, speakers and internet connection. I would also open some of the office applications to make sure they work correctly.**
- **I would check the computer's temperatures and make sure there are no overheating problems or strange noises.**
- **Finally, I would make sure the workstation is clean, the cables are properly connected, and everything is ready for the client to use.**
- **In simple words: I would turn it on, test the hardware and software, install updates, check the connections and make sure everything is working before handing it over.**

Section – B

Task 1: Component Identification & Functions

Component	Function	Make/Model or Specification
Motherboard	The motherboard is the main circuit board of the computer. It connects the CPU, RAM, storage, GPU and other components so they can communicate with each other.	Make/Model: ________
CPU (Processor)	The CPU is the main processing unit. It performs calculations, processes instructions and controls many operations of the computer.	Make/Model: ________
RAM	RAM is temporary memory used by the computer to store data and programs that are currently being used. More RAM can help the computer handle multiple tasks smoothly.	Capacity/Type: ________
GPU	The GPU processes graphics and sends the video output to the monitor. It is especially important for gaming, video editing and other graphics-related tasks.	Make/Model: ________
Storage Device	Storage is used to permanently save the operating system, applications and personal files. Common types are HDD, SATA SSD and NVMe SSD.	Type/Capacity: ________
Cooling System	The cooling system removes heat produced by the CPU and other	Type/Model: ________

	components. It helps prevent overheating and keeps the system working properly.	
PSU	The Power Supply Unit provides electrical power to the motherboard, CPU, GPU, storage and other components.	Wattage: ________

Task 2: Motherboard Ports & Socket Identification

Objective

The purpose of this task is to identify the physical layout of the motherboard, its ports, expansion slots and CPU socket.

Major Motherboard Ports and Connectors

Port/Connector	Purpose
CPU Socket	Holds and connects the CPU to the motherboard.
RAM Slots (DIMM Slots)	Used to install RAM modules.
24-pin ATX Power Connector	Provides main power from the PSU to the motherboard.
CPU Power Connector (4/8-pin)	Provides power to the CPU.
SATA Ports	Used to connect SATA HDDs and SATA SSDs.
M.2 Slot	Used for M.2 SSDs, especially NVMe SSDs.
PCIe x16 Slot	Mainly used for installing a dedicated graphics card.
PCIe x1 Slot	Used for smaller expansion cards such as network or sound cards.

USB Headers	**Connect the computer case's front USB ports to the motherboard.**
Front Panel Header	**Connects the case power button, reset button and indicator LEDs.**
Fan Headers	**Used to connect CPU and case cooling fans.**
Audio Header	**Connects the front audio ports of the computer case.**
CMOS Battery	**Helps maintain BIOS/UEFI settings and system time when the computer is turned off.**
Rear USB Ports	**Used to connect external USB devices such as keyboards, mice and storage devices.**
Ethernet/LAN Port	**Used to connect the computer to a wired network.**
Audio Ports	**Used to connect speakers, headphones and microphones.**
Display Ports	**Used to connect a monitor when supported by the motherboard/CPU.**

CPU Socket Identification

The CPU socket is the part of the motherboard where the processor is installed. The exact socket depends on the motherboard and CPU generation.

For example, an AM5 socket is designed for compatible AMD Ryzen processors, while an LGA 1700 socket supports compatible Intel processor families.

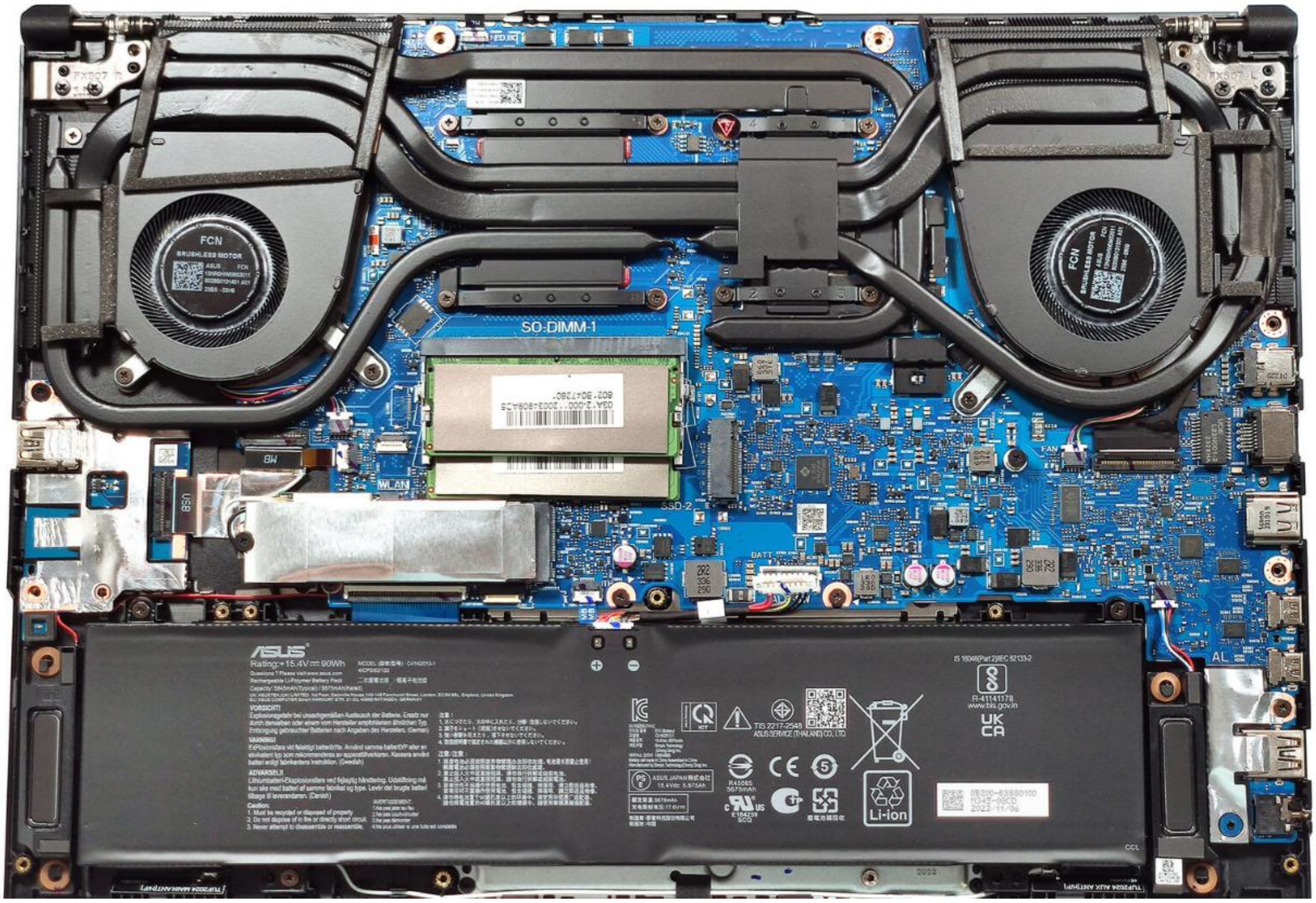
Expansion Slots

The motherboard contains expansion slots that allow additional hardware to be installed.

- **PCIe x16: Usually used for a dedicated graphics card.**

- **PCIe x1: Used for smaller expansion cards.**
- **M.2: Used for compatible M.2 SSDs and, on some boards, other M.2 devices.**
- **SATA connectors: Used to connect SATA storage devices.**

Image of motherboard :



Task 3: RAM & Storage Installation

Objective

The purpose of this task is to safely remove and reinstall RAM and a storage device while following proper safety procedures.

ESD and Safety Precautions

Before working inside the computer, I followed these precautions:

1. **I shut down the computer completely.**
2. **I disconnected the power cable.**
3. **I removed other connected cables where necessary.**
4. **I made sure my hands were clean and dry.**
5. **I took ESD precautions to protect the components from static electricity.**
6. **I handled RAM and storage devices carefully and held them by their edges.**
7. **I did not touch the electrical contacts of the components.**

Removing the RAM

1. **I turned off the computer and disconnected the power.**
2. **I opened the computer case.**
3. **I located the RAM module on the motherboard.**
4. **I opened the locking clips on the RAM slot.**
5. **I carefully removed the RAM module by holding its edges.**
6. **I kept the RAM module in a safe place.**

Reinstalling the RAM

1. I checked the position of the notch on the RAM module.
2. I matched the notch with the slot on the motherboard.
3. I placed the RAM into the correct slot.
4. I pressed it down evenly until the locking clips secured it.
5. I checked that the RAM was properly seated.

Removing and Reinstalling Storage

First, I identified the type of storage device installed in the computer.

Storage Type: SATA SSD / HDD / NVMe SSD

If SATA Storage Was Used

1. I disconnected the SATA data cable.
2. I disconnected the SATA power cable.
3. I removed the drive from its mounting position.
4. I reinstalled the drive securely.
5. I connected the SATA data and power cables again.

If NVMe Storage Was Used

1. I located the M.2 slot on the motherboard.
2. I removed the M.2 retaining screw.
3. I carefully removed the NVMe SSD.
4. I placed the SSD back into the M.2 slot at an angle.
5. I pressed it down gently and secured it with the retaining screw.

Verification

After reinstalling the components, I powered on the computer and checked whether:

- The RAM was detected correctly.

- The storage device was detected correctly.
- The computer started normally.
- There were no warning messages related to RAM or storage.

Task 4: Full Desktop PC Assembly

Objective

The purpose of this task is to assemble a complete desktop computer using the provided components and verify that the system starts successfully.

Components Used

- **Motherboard**
- **CPU**
- **CPU cooling system**
- **RAM**
- **Storage device**
- **GPU**
- **PSU**
- **Computer case**
- **Necessary power and data cables**

Safety Precautions

Before starting the assembly, I followed these safety steps:

1. **I switched off and disconnected the PSU from power.**
2. **I worked on a clean and suitable surface.**
3. **I followed ESD precautions.**
4. **I handled all components carefully.**

5. I did not force any component into a slot.
6. I checked the connectors before connecting cables.
7. I kept screws and small parts organized.

Assembly Procedure

Step 1: Prepare the Motherboard

I placed the motherboard on a safe, non-conductive surface and identified the CPU socket, RAM slots, M.2 slots, PCIe slots and power connectors.

Step 2: Install the CPU

I opened the CPU socket carefully and aligned the CPU with the marking on the socket. I placed the CPU gently into the socket and secured the socket mechanism.

Step 3: Install the CPU Cooler

I installed the CPU cooling system according to the cooler's mounting method. I connected the CPU fan cable to the appropriate CPU fan header on the motherboard.

Step 4: Install the RAM

I installed the RAM modules into the correct DIMM slots. I matched the notch on the RAM with the slot and pressed the module down until it was properly locked.

Step 5: Install Storage

I installed the storage device. If an NVMe SSD was used, I installed it into the M.2 slot and secured it with the screw. If SATA storage was used, I connected the SATA data and power cables.

Step 6: Install the Motherboard in the Case

I carefully placed the motherboard inside the computer case and aligned it with the case mounting points. I secured it using the appropriate screws.

Step 7: Install the PSU

I installed the power supply unit in the case and secured it properly.

Step 8: Install the GPU

I installed the graphics card into the appropriate PCIe x16 slot. I secured the GPU to the case and connected the required power cable if needed.

Step 9: Connect Power Cables

I connected the required power cables, including:

- 24-pin motherboard power cable
- CPU power cable
- GPU power cable, if required
- SATA power cable, if SATA storage was used

Step 10: Connect Front Panel and Other Cables

I connected the case's front-panel connectors, USB headers, audio header and case fan cables to the correct motherboard headers.

Step 11: Final Inspection

Before powering on the system, I checked that:

- RAM was properly seated.
- CPU cooler was properly installed.
- GPU was properly seated.
- Storage was connected.
- Power cables were connected correctly.
- No loose screws or cables were inside the case.
- All fans could rotate freely.

Step 12: Power On and POST Test

I connected the power cable and switched on the PSU. I then pressed the computer's power button.

The system successfully started and performed POST (Power-On Self-Test).

I checked that:

- **The system powered on.**
- **Fans started running.**
- **The display appeared on the monitor.**
- **The system entered BIOS/UEFI or started the operating system.**
- **RAM and storage were detected.**

POST Result: Successful / Not Successful

RAM Detected: ___________________

Storage Detected: ___________________

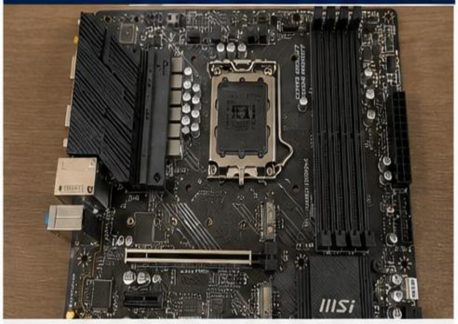
GPU Detected: ___________________

Documentation

I documented the assembly process using photographs/diagrams at important stages:

1. **Motherboard before assembly**
2. **CPU installation**
3. **RAM installation**
4. **Storage installation**
5. **Motherboard installed in case**
6. **GPU installation**
7. **PSU and cable connections**
8. **Completed PC assembly**
9. **System powered on and POST completed**

1. Motherboard Before Assembly

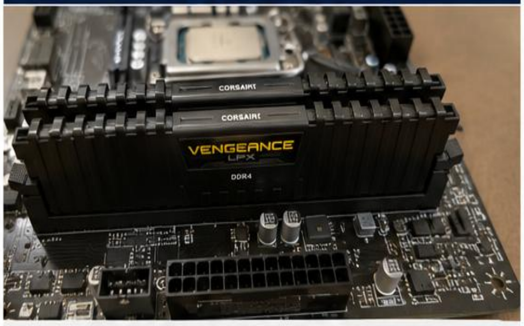


Motherboard before installing any components.

2. CPU Installation

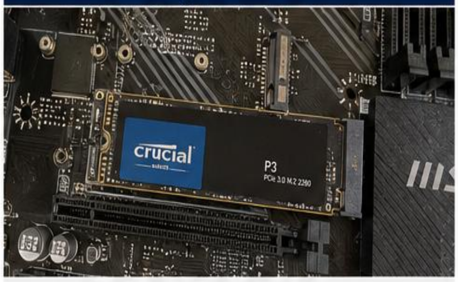
CPU installed correctly in the motherboard socket.

3. RAM Installation



RAM module installed in the DIMM slot.

4. Storage Installation



NVMe SSD installed in the M.2 slot.

5. Motherboard Installed in Case

Motherboard installed and secured inside the case.

6. GPU Installation

Graphics card installed in the PCIe x16 slot.

7. PSU and Cable Connections

PSU installed and all required cables connected.

8. Completed PC Assembly

Complete PC assembly with all components installed.

9. System Powered On and POST Completed

System powered on successfully and POST completed.

Section – C

1. Three Components to Check First

1. PSU

- **Check whether the PSU is switched ON.**
- **Check the power cable.**
- **Check the 24-pin motherboard and CPU power cables.**
- **A faulty or disconnected PSU can prevent the PC from starting.**

2. Motherboard

- **Check motherboard power connections.**
- **Check CPU installation and diagnostic LEDs.**
- **Make sure the motherboard and CPU are compatible.**

3. RAM

- **Check whether RAM is properly seated.**
- **Check the RAM slot and locking clips.**
- **BIOS should show the correct RAM capacity.**

2. Motherboard–CPU Compatibility

The CPU and motherboard must have a compatible socket and supported CPU generation.

For example, our Intel Core i5-12400F requires an LGA 1700-compatible motherboard. If the motherboard does not support the CPU, the computer may not POST or display anything.

3. Checking RAM and Storage

RAM

1. **Turn off the PC and disconnect power.**
2. **Check that RAM is fully inserted.**
3. **Make sure the clips are locked.**
4. **Check BIOS to confirm the correct RAM capacity.**

Storage

1. **Check the NVMe SSD is properly fitted in the M.2 slot.**
2. **For SATA drives, check the SATA data and power cables.**
3. **Check BIOS to confirm that the storage device is detected.**

4. Installation Checklist

- **Follow ESD and safety precautions.**
- **Check CPU and motherboard compatibility.**
- **Install RAM and storage correctly.**
- **Connect all motherboard and PSU cables.**

- **Check all components before closing the case.**
- **Power on and confirm successful POST.**

5. Why Use SSD Instead of HDD?

SSDs are better for modern office PCs because they:

- **Boot the computer faster.**
- **Open applications quickly.**
- **Transfer files faster.**
- **Make less noise.**
- **Have no moving mechanical parts.**

Therefore, SSDs are better

Section D — Correcting AI

Step 1: Exact AI Prompt

"Give me a simple step-by-step checklist for diagnosing a desktop PC that fails to boot after assembly. Include PSU, motherboard, RAM, storage, GPU, BIOS/POST and ESD safety. Apply it to the ABC Solutions scenario."

AI's Original Checklist

1. **Check PSU and power cables.**
2. **Check motherboard and CPU.**
3. **Check RAM.**
4. **Check GPU.**

5. Check storage.
6. Check BIOS/POST.

My Corrected Checklist

1. Follow ESD safety and disconnect power.
2. Check PSU and power cables.
3. Check motherboard and CPU compatibility.
4. Check RAM installation.
5. Check GPU and display.
6. Check storage.
7. Check BIOS/POST.
8. Check software only after hardware passes POST.

What I Changed

The AI checklist did not clearly put ESD safety first. I added it as the first step because touching components without proper safety can damage hardware. I also placed software checks at the end because hardware and POST should be checked first.